

Introduction

This project reviews a simple login system and identifies security vulnerabilities

Code Reviewed

```
username = input("Enter Username: ")
password = input("Enter Password: ")

if len(username) < 3:
    print("Invalid Username")

elif len(password) < 6:
    print("Password must be at least 6 characters")

elif username == "admin" and password == "1234hs":
    print("Login Successful")

else:
    print("Login Failed")
```

Vulnerabilities Found are :

1. No Account Lockout Mechanism

The program allows unlimited login attempts.

Risk: Attackers can perform brute-force attacks.

2. No Multi-Factor Authentication (MFA)

Authentication relies only on a username and password.

Risk: A compromised password can lead to unauthorized access

3. No Secure Credential Storage

Credentials are not stored in a database or secure vault.

Risk: Sensitive information can be easily exposed.

1. Hardcoded Credentials

The username and password are directly written in the source code.

Recommendations

1. Use password hashing
2. Store credentials securely in database
3. Validate user input
4. Implement account lockout after multiple failed login attempts
And also Implement strong password policy

Conclusion

The application contains multiple security vulnerabilities. Following secure coding practices can improve security and reduce risks.